

# Day 19

## UVM: Universal Verification Language

**The real power of UVM:** The ability to create completely different test scenarios by only changing the test class, leaving everything else identical.

Without UVM, the entire testbench should be modified, but here, just by adding a couple of test classes and sequences, we can test specifically

**To the same program as previous day, adding 2 extra sequences and test classes**

### // Sequence 2 — overflow test

// Force a+b to exceed 4 bits (sum > 15)

```
class adder_overflow_seq extends uvm_sequence #(adder_seq_item);
```

```
`uvm_object_utils(adder_overflow_seq)
```

```
function new(string name = "adder_overflow_seq");
```

```
    super.new(name);
```

```
endfunction
```

```
task body();
```

```
    adder_seq_item item;
```

```
    repeat(10) begin
```

```
        item = adder_seq_item::type_id::create("item");
```

```
        start_item(item);
```

```
        // Constrain so sum always overflows 4 bits
```

```
        assert(item.randomize() with {
```

```
            a + b > 15; // force overflow
```

```
        });
```

```
        finish_item(item);
```

```
    end
```

```
endtask
```

```
endclass
```

### // Sequence 3 — corner case: a=0 or b=0

```
class adder_zero_seq extends uvm_sequence #(adder_seq_item);
```

```
`uvm_object_utils(adder_zero_seq)
```

```
function new(string name = "adder_zero_seq");
```

```
    super.new(name);
```

```
endfunction
```

```
task body();
```

```
    adder_seq_item item;
```

```
    // Test a=0, all values of b
```

```
    repeat(5) begin
```

```
        item = adder_seq_item::type_id::create("item");
```

```
        start_item(item);
```

```
        assert(item.randomize() with { a == 0; });
```

```
        finish_item(item);
```

```
    end
```

```
    // Test b=0, all values of a
```

```
    repeat(5) begin
```

```

        item = adder_seq_item::type_id::create("item");
        start_item(item);
        assert(item.randomize() with { b == 0; });
        finish_item(item);
    end
endtask
endclass

```

## // Test 2 — overflow test

```

class adder_overflow_test extends uvm_test;
    `uvm_component_utils(adder_overflow_test)

    adder_env env;

    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        env = adder_env::type_id::create("env", this);
    endfunction

    task run_phase(uvm_phase phase);
        adder_overflow_seq seq;
        phase.raise_objection(this);
        `uvm_info("TEST", "Running overflow test", UVM_NONE)
        seq = adder_overflow_seq::type_id::create("seq");
        seq.start(env.agent.seqr);
        phase.drop_objection(this);
    endtask
endclass

```

## // Test 3 — zero test

```

class adder_zero_test extends uvm_test;
    `uvm_component_utils(adder_zero_test)

    adder_env env;

    function new(string name, uvm_component parent);
        super.new(name, parent);
    endfunction

    function void build_phase(uvm_phase phase);
        super.build_phase(phase);
        env = adder_env::type_id::create("env", this);
    endfunction

    task run_phase(uvm_phase phase);
        adder_zero_seq seq;
        phase.raise_objection(this);
        `uvm_info("TEST", "Running zero test", UVM_NONE)
        seq = adder_zero_seq::type_id::create("seq");
        seq.start(env.agent.seqr);
        phase.drop_objection(this);
    endtask
endclass

```

## Changing the Run option in EDA playground:

+UVM\_TESTNAME=adderr\_overflow\_test

+UVM\_TESTNAME=adderr\_zero\_test

### Output for adderr\_overflow\_test:

```
UVM_INFO @ 0: reporter [RNTST] Running test adderr_overflow_test...
UVM_INFO testbench.sv(265) @ 0: uvm_test_top [TEST] Running overflow test
UVM_INFO testbench.sv(171) @ 10: uvm_test_top.env.sb [SB] PASS: a=9 b=14 sum=23 | expected=23
UVM_INFO testbench.sv(147) @ 10: uvm_test_top.env.agent.mon [MON] Observed: a=9 b=14 sum=23
UVM_INFO testbench.sv(115) @ 10: uvm_test_top.env.agent.drv [DRV] Driving a=9 b=14
UVM_INFO testbench.sv(171) @ 20: uvm_test_top.env.sb [SB] PASS: a=5 b=11 sum=16 | expected=16
UVM_INFO testbench.sv(147) @ 20: uvm_test_top.env.agent.mon [MON] Observed: a=5 b=11 sum=16
UVM_INFO testbench.sv(115) @ 20: uvm_test_top.env.agent.drv [DRV] Driving a=5 b=11
UVM_INFO testbench.sv(171) @ 30: uvm_test_top.env.sb [SB] PASS: a=14 b=7 sum=21 | expected=21
UVM_INFO testbench.sv(147) @ 30: uvm_test_top.env.agent.mon [MON] Observed: a=14 b=7 sum=21
UVM_INFO testbench.sv(115) @ 30: uvm_test_top.env.agent.drv [DRV] Driving a=14 b=7
UVM_INFO testbench.sv(171) @ 40: uvm_test_top.env.sb [SB] PASS: a=8 b=12 sum=20 | expected=20
UVM_INFO testbench.sv(147) @ 40: uvm_test_top.env.agent.mon [MON] Observed: a=8 b=12 sum=20
UVM_INFO testbench.sv(115) @ 40: uvm_test_top.env.agent.drv [DRV] Driving a=8 b=12
UVM_INFO testbench.sv(171) @ 50: uvm_test_top.env.sb [SB] PASS: a=14 b=3 sum=17 | expected=17
UVM_INFO testbench.sv(147) @ 50: uvm_test_top.env.agent.mon [MON] Observed: a=14 b=3 sum=17
UVM_INFO testbench.sv(115) @ 50: uvm_test_top.env.agent.drv [DRV] Driving a=14 b=3
UVM_INFO testbench.sv(171) @ 60: uvm_test_top.env.sb [SB] PASS: a=10 b=6 sum=16 | expected=16
UVM_INFO testbench.sv(147) @ 60: uvm_test_top.env.agent.mon [MON] Observed: a=10 b=6 sum=16
UVM_INFO testbench.sv(115) @ 60: uvm_test_top.env.agent.drv [DRV] Driving a=10 b=6
UVM_INFO testbench.sv(171) @ 70: uvm_test_top.env.sb [SB] PASS: a=14 b=13 sum=27 | expected=27
UVM_INFO testbench.sv(147) @ 70: uvm_test_top.env.agent.mon [MON] Observed: a=14 b=13 sum=27
UVM_INFO testbench.sv(115) @ 70: uvm_test_top.env.agent.drv [DRV] Driving a=14 b=13
UVM_INFO testbench.sv(171) @ 80: uvm_test_top.env.sb [SB] PASS: a=13 b=5 sum=18 | expected=18
UVM_INFO testbench.sv(147) @ 80: uvm_test_top.env.agent.mon [MON] Observed: a=13 b=5 sum=18
UVM_INFO testbench.sv(115) @ 80: uvm_test_top.env.agent.drv [DRV] Driving a=13 b=5
UVM_INFO testbench.sv(171) @ 90: uvm_test_top.env.sb [SB] PASS: a=13 b=3 sum=16 | expected=16
UVM_INFO testbench.sv(147) @ 90: uvm_test_top.env.agent.mon [MON] Observed: a=13 b=3 sum=16
UVM_INFO testbench.sv(115) @ 90: uvm_test_top.env.agent.drv [DRV] Driving a=13 b=3
UVM_INFO testbench.sv(171) @ 100: uvm_test_top.env.sb [SB] PASS: a=15 b=14 sum=29 | expected=29
UVM_INFO testbench.sv(147) @ 100: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=14 sum=29
UVM_INFO testbench.sv(115) @ 100: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=14
UVM_INFO /xcelium25.03/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_objection.svh(1268) @
100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO testbench.sv(185) @ 100: uvm_test_top.env.sb [SB] === SCOREBOARD: 10 PASS 0 FAIL ===
```

--- UVM Report catcher Summary ---

```
Number of demoted UVM_FATAL reports : 0
Number of demoted UVM_ERROR reports : 0
Number of demoted UVM_WARNING reports: 0
Number of caught UVM_FATAL reports : 0
Number of caught UVM_ERROR reports : 0
Number of caught UVM_WARNING reports : 0
```

--- UVM Report Summary ---

\*\* Report counts by severity

```
UVM_INFO : 34
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
```

\*\* Report counts by id

```
[DRV] 10
[MON] 10
[RNTST] 1
[SB] 11
[TEST] 1
[TEST_DONE] 1
Simulation complete via $finish(1) at time 100 NS + 49
```

### Output for adderr\_zero\_test:

```
UVM_INFO @ 0: reporter [RNTST] Running test adderr_zero_test...
UVM_INFO testbench.sv(290) @ 0: uvm_test_top [TEST] Running zero test
UVM_INFO testbench.sv(171) @ 10: uvm_test_top.env.sb [SB] PASS: a=0 b=8 sum=8 | expected=8
UVM_INFO testbench.sv(147) @ 10: uvm_test_top.env.agent.mon [MON] Observed: a=0 b=8 sum=8
UVM_INFO testbench.sv(115) @ 10: uvm_test_top.env.agent.drv [DRV] Driving a=0 b=8
UVM_INFO testbench.sv(171) @ 20: uvm_test_top.env.sb [SB] PASS: a=0 b=5 sum=5 | expected=5
UVM_INFO testbench.sv(147) @ 20: uvm_test_top.env.agent.mon [MON] Observed: a=0 b=5 sum=5
UVM_INFO testbench.sv(115) @ 20: uvm_test_top.env.agent.drv [DRV] Driving a=0 b=5
UVM_INFO testbench.sv(171) @ 30: uvm_test_top.env.sb [SB] PASS: a=0 b=15 sum=15 | expected=15
UVM_INFO testbench.sv(147) @ 30: uvm_test_top.env.agent.mon [MON] Observed: a=0 b=15 sum=15
UVM_INFO testbench.sv(115) @ 30: uvm_test_top.env.agent.drv [DRV] Driving a=0 b=15
UVM_INFO testbench.sv(171) @ 40: uvm_test_top.env.sb [SB] PASS: a=0 b=7 sum=7 | expected=7
UVM_INFO testbench.sv(147) @ 40: uvm_test_top.env.agent.mon [MON] Observed: a=0 b=7 sum=7
```

```

UVM_INFO testbench.sv(115) @ 40: uvm_test_top.env.agent.drv [DRV] Driving a=0 b=7
UVM_INFO testbench.sv(171) @ 50: uvm_test_top.env.sb [SB] PASS: a=0 b=13 sum=13 | expected=13
UVM_INFO testbench.sv(147) @ 50: uvm_test_top.env.agent.mon [MON] Observed: a=0 b=13 sum=13
UVM_INFO testbench.sv(115) @ 50: uvm_test_top.env.agent.drv [DRV] Driving a=0 b=13
UVM_INFO testbench.sv(171) @ 60: uvm_test_top.env.sb [SB] PASS: a=15 b=0 sum=15 | expected=15
UVM_INFO testbench.sv(147) @ 60: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=0 sum=15
UVM_INFO testbench.sv(115) @ 60: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=0
UVM_INFO testbench.sv(171) @ 70: uvm_test_top.env.sb [SB] PASS: a=15 b=0 sum=15 | expected=15
UVM_INFO testbench.sv(147) @ 70: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=0 sum=15
UVM_INFO testbench.sv(115) @ 70: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=0
UVM_INFO testbench.sv(171) @ 80: uvm_test_top.env.sb [SB] PASS: a=12 b=0 sum=12 | expected=12
UVM_INFO testbench.sv(147) @ 80: uvm_test_top.env.agent.mon [MON] Observed: a=12 b=0 sum=12
UVM_INFO testbench.sv(115) @ 80: uvm_test_top.env.agent.drv [DRV] Driving a=12 b=0
UVM_INFO testbench.sv(171) @ 90: uvm_test_top.env.sb [SB] PASS: a=6 b=0 sum=6 | expected=6
UVM_INFO testbench.sv(147) @ 90: uvm_test_top.env.agent.mon [MON] Observed: a=6 b=0 sum=6
UVM_INFO testbench.sv(115) @ 90: uvm_test_top.env.agent.drv [DRV] Driving a=6 b=0
UVM_INFO testbench.sv(171) @ 100: uvm_test_top.env.sb [SB] PASS: a=14 b=0 sum=14 | expected=14
UVM_INFO testbench.sv(147) @ 100: uvm_test_top.env.agent.mon [MON] Observed: a=14 b=0 sum=14
UVM_INFO testbench.sv(115) @ 100: uvm_test_top.env.agent.drv [DRV] Driving a=14 b=0
UVM_INFO /xcelium25.03/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_objection.svh(1268) @
100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO testbench.sv(185) @ 100: uvm_test_top.env.sb [SB] === SCOREBOARD: 10 PASS 0 FAIL ===

```

--- UVM Report catcher Summary ---

```

Number of demoted UVM_FATAL reports : 0
Number of demoted UVM_ERROR reports : 0
Number of demoted UVM_WARNING reports : 0
Number of caught UVM_FATAL reports : 0
Number of caught UVM_ERROR reports : 0
Number of caught UVM_WARNING reports : 0

```

--- UVM Report Summary ---

```

** Report counts by severity
UVM_INFO : 34
UVM_WARNING : 0
UVM_ERROR : 0
UVM_FATAL : 0
** Report counts by id
[DRV] 10
[MON] 10
[RNTST] 1
[SB] 11
[TEST] 1
[TEST_DONE] 1
Simulation complete via $finish(1) at time 100 NS + 49

```

### Deliberately causing an error in the design to check:

```

UVM_INFO @ 0: reporter [RNTST] Running test adder_test...
UVM_ERROR testbench.sv(177) @ 10: uvm_test_top.env.sb [SB] FAIL: a=8 b=13 sum=27 | expected=21
got=27
UVM_INFO testbench.sv(147) @ 10: uvm_test_top.env.agent.mon [MON] Observed: a=8 b=13 sum=27
UVM_INFO testbench.sv(115) @ 10: uvm_test_top.env.agent.drv [DRV] Driving a=8 b=13
UVM_ERROR testbench.sv(177) @ 20: uvm_test_top.env.sb [SB] FAIL: a=5 b=7 sum=30 | expected=12
got=30
UVM_INFO testbench.sv(147) @ 20: uvm_test_top.env.agent.mon [MON] Observed: a=5 b=7 sum=30
UVM_INFO testbench.sv(115) @ 20: uvm_test_top.env.agent.drv [DRV] Driving a=5 b=7
UVM_ERROR testbench.sv(177) @ 30: uvm_test_top.env.sb [SB] FAIL: a=15 b=13 sum=2 | expected=28
got=2
UVM_INFO testbench.sv(147) @ 30: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=13 sum=2
UVM_INFO testbench.sv(115) @ 30: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=13
UVM_ERROR testbench.sv(177) @ 40: uvm_test_top.env.sb [SB] FAIL: a=7 b=11 sum=28 | expected=18
got=28
UVM_INFO testbench.sv(147) @ 40: uvm_test_top.env.agent.mon [MON] Observed: a=7 b=11 sum=28
UVM_INFO testbench.sv(115) @ 40: uvm_test_top.env.agent.drv [DRV] Driving a=7 b=11
UVM_ERROR testbench.sv(177) @ 50: uvm_test_top.env.sb [SB] FAIL: a=13 b=2 sum=11 | expected=15
got=11
UVM_INFO testbench.sv(147) @ 50: uvm_test_top.env.agent.mon [MON] Observed: a=13 b=2 sum=11
UVM_INFO testbench.sv(115) @ 50: uvm_test_top.env.agent.drv [DRV] Driving a=13 b=2
UVM_ERROR testbench.sv(177) @ 60: uvm_test_top.env.sb [SB] FAIL: a=15 b=2 sum=13 | expected=17
got=13
UVM_INFO testbench.sv(147) @ 60: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=2 sum=13
UVM_INFO testbench.sv(115) @ 60: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=2
UVM_ERROR testbench.sv(177) @ 70: uvm_test_top.env.sb [SB] FAIL: a=15 b=13 sum=2 | expected=28
got=2
UVM_INFO testbench.sv(147) @ 70: uvm_test_top.env.agent.mon [MON] Observed: a=15 b=13 sum=2
UVM_INFO testbench.sv(115) @ 70: uvm_test_top.env.agent.drv [DRV] Driving a=15 b=13
UVM_ERROR testbench.sv(177) @ 80: uvm_test_top.env.sb [SB] FAIL: a=12 b=4 sum=8 | expected=16
got=8
UVM_INFO testbench.sv(147) @ 80: uvm_test_top.env.agent.mon [MON] Observed: a=12 b=4 sum=8

```

```

UVM_INFO testbench.sv(115) @ 80: uvm_test_top.env.agent.drv [DRV] Driving a=12 b=4
UVM_ERROR testbench.sv(177) @ 90: uvm_test_top.env.sb [SB] FAIL: a=6 b=5 sum=1 | expected=11
got=1
UVM_INFO testbench.sv(147) @ 90: uvm_test_top.env.agent.mon [MON] Observed: a=6 b=5 sum=1
UVM_INFO testbench.sv(115) @ 90: uvm_test_top.env.agent.drv [DRV] Driving a=6 b=5
UVM_ERROR testbench.sv(177) @ 100: uvm_test_top.env.sb [SB] FAIL: a=14 b=13 sum=1 | expected=27
got=1
UVM_INFO testbench.sv(147) @ 100: uvm_test_top.env.agent.mon [MON] Observed: a=14 b=13 sum=1
UVM_INFO testbench.sv(115) @ 100: uvm_test_top.env.agent.drv [DRV] Driving a=14 b=13
UVM_INFO /xcelium25.03/tools/methodology/UVM/CDNS-1.1d/sv/src/base/uvm_objection.svh(1268) @
100: reporter [TEST_DONE] 'run' phase is ready to proceed to the 'extract' phase
UVM_INFO testbench.sv(185) @ 100: uvm_test_top.env.sb [SB] === SCOREBOARD: 0 PASS 10 FAIL ===

```

--- UVM Report catcher Summary ---

```

Number of demoted UVM_FATAL reports : 0
Number of demoted UVM_ERROR reports : 0
Number of demoted UVM_WARNING reports: 0
Number of caught UVM_FATAL reports : 0
Number of caught UVM_ERROR reports : 0
Number of caught UVM_WARNING reports : 0

```

--- UVM Report Summary ---

```

** Report counts by severity
UVM_INFO : 23
UVM_WARNING : 0
UVM_ERROR : 10
UVM_FATAL : 0
** Report counts by id
[DRV] 10
[MON] 10
[RNTST] 1
[SB] 11
[TEST_DONE] 1
Simulation complete via $finish(1) at time 100 NS + 49

```

**We can see all cases failed, just because I changed sum = a-b in the top module**